

LARKHILL, United Kingdom

WMO: 037430

Lat: 51.2017N

Lon: 1.8058W

Elev: 436

StdP: 14.47

Time Zone: 0.00 (EUW)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.5	27.9	20.8	15.8	29.7	23.3	17.9	31.1	29.4	48.1	26.2	47.6	6.8	40	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	79.3	64.9	75.5	63.1	72.4	61.7	66.4	75.7	64.6	72.6	63.0	69.8	7.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	87.9	69.1	61.7	83.4	67.1	60.4	79.6	65.5	31.3	75.9	29.9	72.7	28.7	70.1	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	20.2	17.8	DB	20.2	84.2	3.3	3.1	17.8	86.5	15.8	88.3	14.0	90.0	11.6	92.3	(4)
(5)			WB	19.6	70.5	3.3	3.9	17.2	73.3	15.2	75.6	13.4	77.8	11.0	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	40.3	40.9	43.7	47.8	53.1	58.3	62.1	61.8	57.5	52.1	45.1	41.0		
	DBStd	9.46	6.01	5.65	5.22	5.23	5.18	4.43	4.41	4.18	4.11	5.02	5.53	6.56		
	HDD50	1367	302	257	202	99	25	1	0	0	1	35	162	283		
	HDD65	5391	765	676	660	516	368	206	114	118	228	401	596	743		
	CDD50	1504	1	1	7	33	123	250	375	367	226	100	16	5		
	CDD65	53	0	0	0	0	0	6	25	20	2	0	0	0		
	CDH74	467	0	0	0	1	8	67	224	153	12	2	0	0		
	CDH80	71	0	0	0	0	0	8	36	27	0	0	0	0		
(14) Wind	WSAvg	8.4	9.6	9.8	9.4	8.7	8.4	7.7	7.3	7.1	7.5	8.4	8.3	9.0		
(15) Precipitation	PrecAvg	30.2	3.3	2.2	2.6	1.7	2.0	2.0	1.8	2.2	2.6	3.0	3.0	3.6		
	PrecMax	41.7	6.0	6.4	5.4	3.6	4.3	5.0	3.6	5.4	7.0	9.6	7.6	6.7		
	PrecMin	22.3	0.3	0.2	0.5	0.0	0.4	0.2	0.6	0.5	0.8	0.1	1.2	0.7		
	PrecStd	4.8	1.6	1.6	1.2	0.9	1.2	1.2	0.9	1.4	1.6	2.0	1.7	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	54.9	62.1	70.8	75.0	80.6	83.9	83.4	75.6	67.3	59.3	56.1		
		MCWB	51.8	50.7	51.9	57.1	61.9	64.9	66.4	66.3	63.2	61.0	56.8	53.6		
	2%	DB	52.6	52.4	57.0	64.3	71.0	75.3	80.0	78.2	71.4	63.2	57.0	54.2		
		MCWB	50.6	49.6	49.7	54.1	59.9	62.7	65.3	64.7	61.7	58.3	54.7	52.3		
	5%	DB	51.0	50.7	54.1	59.8	67.1	71.6	76.1	73.8	68.0	61.2	55.4	52.7		
		MCWB	49.1	48.1	48.8	51.6	58.1	61.1	63.4	62.7	59.7	57.4	52.9	50.8		
	10%	DB	49.2	49.1	51.8	56.7	63.0	68.1	72.2	70.4	65.3	59.5	53.8	50.9		
		MCWB	47.4	46.7	47.8	49.7	55.6	59.2	61.7	61.3	58.4	56.3	51.6	48.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	52.3	53.2	58.6	65.0	67.2	68.7	69.1	65.6	61.9	57.5	54.4		
		MCDB	53.5	54.1	57.9	68.1	71.8	77.6	80.0	79.2	71.9	66.3	58.6	55.3		
	2%	WB	51.2	50.4	51.4	54.9	61.3	64.2	66.3	66.3	63.2	59.9	55.2	52.9		
		MCDB	52.4	51.9	55.0	62.8	68.2	72.6	77.2	74.1	68.5	62.1	56.5	54.0		
	5%	WB	49.3	48.7	50.2	52.7	58.9	62.1	64.5	64.3	61.5	58.4	53.6	51.2		
		MCDB	50.7	50.2	53.2	58.3	65.7	69.2	73.4	71.2	65.8	60.4	55.1	52.6		
	10%	WB	47.7	47.1	48.6	50.9	56.7	60.3	62.7	62.7	60.0	57.1	51.8	49.3		
		MCDB	49.1	48.9	51.3	55.5	62.1	66.7	70.3	68.4	63.9	59.0	53.6	50.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.3	12.7	15.1	15.6	15.9	16.3	15.5	14.9	11.9	10.3	9.1		
		MCDBR	10.0	10.8	16.1	19.9	21.3	22.0	23.4	21.2	18.3	12.5	10.0	9.0		
	5% WB	MCWBR	9.8	8.9	10.5	11.6	12.4	11.8	10.9	9.8	9.9	8.6	8.7	8.9		
		MCWBR	10.2	10.5	13.8	17.6	19.1	19.3	20.2	17.8	15.8	10.8	9.8	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.338	0.374	0.395	0.399	0.393	0.388	0.379	0.366	0.351	0.325	0.313		
		taud	2.360	2.374	2.273	2.228	2.254	2.298	2.321	2.367	2.395	2.418	2.406	2.341		
	Ebn at Noon		218	249	259	267	272	273	273	270	260	241	216	201		
		Edh at Noon	20	26	34	40	41	39	38	35	30	24	19	18		
(44) All-Sky Solar Radiation	RadAvg		258	476	838	1305	1584	1742	1647	1372	1029	588	311	202		
	RadStd		21	59	110	140	93	174	147	111	80	50	34	24		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page